

Zintl-phase Mg_3Sb_2 as an emerging infrared photodetector material on silicon

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ABSTRACT

Infrared (IR) photodetectors integrated on Si are in critical demand for Internet of Things (IoT) and AI applications. We repositioned the Zintl-phase compound Mg_3Sb_2 , widely known as a high-performance thermoelectric material, as a candidate for Si-integrated optoelectronics. We demonstrate the first epitaxial growth of Mg_3Sb_2 thin films on Si(001) substrates, revealing a unique c axis orientation with four three-fold symmetric domains. A p–n heterojunction formed between p-type Mg_3Sb_2 and n-type Si exhibited clear rectification. Crucially, the device demonstrated a distinct photoresponse to short-wavelength infrared light (1450 and 1600 nm), where we obtained spectral sensitivity spectra and confirmed a linear dependence of the photocurrent on the incident power at infrared wavelengths. This work unveils Mg_3Sb_2 's untapped optoelectronic properties, establishing $\text{Mg}_3\text{Sb}_2/\text{Si}$ as a promising platform for IR photodetectors.

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Infrared (IR) sensing technology has become indispensable for artificial intelligence (AI), IoT, and autonomous driving applications.^{1–3} Since the early discovery of IR radiation⁴ and thermal detectors,⁵ the field has evolved to high-performance photon detectors. Currently, InGaAs,^{6–9} HgCdTe,^{10–14} and InAsSb^{15,16} define the state-of-the-art. However, these materials typically require expensive, non-scalable substrates and often contain hazardous elements. Furthermore, unlike near-infrared light, which poses retinal risks,¹⁷ short-wavelength infrared (SWIR) is “eye-safe,” enabling higher-power applications. To address these limitations, PbS quantum dots^{18–20} have emerged as candidates but suffer from low chemical stability.^{21–23} In response to the demand for sustainable IR sensing platforms, our group has previously proposed an alternative approach using Mg_2Si .^{24–26} By developing Mg_2Si bulk single-crystal-based photodiodes, we demonstrated that Mg-based compounds can serve as effective, earth-abundant, and environmentally friendly SWIR detection materials. To further advance this technology and fully leverage the mature Si photonics infrastructure, there is an ongoing need to explore material candidates in parallel with other materials under investigation for Si integration, such as GeSn.^{27–29}

We focus on the Zintl-phase compound Mg_3Sb_2 as a promising platform for Si-integrated SWIR photodetectors. Mg_3Sb_2 possesses an indirect bandgap of approximately 0.6 eV,³⁰ making it suitable for

SWIR sensors. A key advantage of this material is its exceptional compositional tunability; the bandgap can be continuously tuned from 0.6 eV down to nearly 0 eV (Ref. 30) by alloying with bismuth (Bi) to form $\text{Mg}_3\text{Sb}_{2-x}\text{Bi}_x$, enabling precise optimization for eye-safe detection wavelengths. Furthermore, Mg_3Sb_2 features a unique band structure with a high conduction band valley degeneracy of 6,^{30,31} leading to a large density of states (DOS) that is expected to yield a strong photoresponse. However, its potential as a photodetector material has been previously overlooked. While its thermoelectric properties are well-documented,^{30–46} reports on its photoresponse properties are conspicuously absent from the literature. This Letter reports the SWIR photoresponse of Mg_3Sb_2 thin films grown on Si substrates, demonstrating their strong potential for this application.

Our group has previously reported the epitaxial growth of Mg_3Sb_2 and Mg_3Bi_2 thin films on c-plane Al_2O_3 substrates using the molecular beam epitaxy (MBE) method.^{46,47} Both films exhibited p-type conductivity despite being in an undoped state. This epitaxial growth is enabled by the fact that both Mg_3Sb_2 or Mg_3Bi_2 and the c-plane Al_2O_3 substrate belong to the trigonal crystal system and have a small lattice mismatch of 2%–4.2%. In contrast, the epitaxial relationship between Mg_3Sb_2 and an Si substrate has not been clarified because of the differences in their crystal systems. Since the fundamental physical parameters of Mg_3Sb_2 and Si, including crystal structure, lattice

TABLE I. Material parameters of Si and Mg₃Sb₂.

	Crystal structure	Lattice constant (Å)	Thermal expansion coefficient (K ⁻¹)
Si	Cubic	$a = 5.43$	3×10^{-6}
Mg ₃ Sb ₂	Trigonal	$a = 4.53$, $c = 7.24$ (Ref. 48)	50×10^{-6} (Ref. 49)

constants,⁴⁸ and thermal expansion coefficients⁴⁹ are quite different, as shown in Table I, it is unclear what crystal orientation the epitaxial thin film of Mg₃Sb₂ on a Si substrate will have.

Nevertheless, the epitaxial growth has been reported for similar hexagonal materials such as GaN^{50,51} and AlN⁵² on Si (111) or ZnO⁵³ on Si (001) by utilizing specific plane orientation relationships. This suggests that the epitaxial growth of Mg₃Sb₂ on Si substrates is feasible.

In this study, Mg₃Sb₂ was epitaxially grown on a Si substrate using the MBE method, and its crystal structure and IR photodetection properties were investigated. We demonstrated *c* axis-oriented epitaxial growth and observed the IR photoresponse.

The Mg₃Sb₂ thin films were epitaxially grown on n-Si (001) substrates (>1000 Ω cm) using an MBE system (EIKO, EW-100, base pressure $\sim 10^{-8}$ Pa). The Si (001) substrates underwent organic treatment, acid treatment with piranha solution (a mixture of H₂SO₄ and H₂O₂), hydrofluoric acid treatment, and a final acid treatment with piranha solution to achieve chemical cleaning and to form chemical oxide layers. After the Si substrates were introduced into the MBE chamber, Si was deposited at 650 °C to form a 30 nm Si buffer layer, achieving a clean surface with a (2 × 1) surface reconstruction. The Mg₃Sb₂ thin films were grown by simultaneously depositing Mg and Sb with a flux ratio of 4:1 using Knudsen cells. The Mg and Sb raw materials were high purity 5N and 6N, respectively. Samples were prepared by varying the substrate temperature during growth to room temperature (RT), 300, 500, and 550 °C.

The surface structures of the thin films before and after growth were investigated *in situ* using reflection high-energy electron diffraction (RHEED). The crystal structure was characterized by x-ray diffraction (XRD) using a Cu Kα source and a Ni Kβ filter (Rigaku, Ultima IV). Cross-sectional specimens for TEM observations were prepared using a focused ion beam (FIB) system (JEOL, JIB-4700F). The cross-sectional crystal structure was further observed using high-resolution transmission electron microscopy (HRTEM) and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) (JEOL, JEM-ARM200F).

To evaluate the photoresponse, two types of device configurations were employed. The initial demonstration was performed using the sample (3 × 5 mm² in size) with manually applied indium (In) top contacts, which was attached to a Cu stage using conductive Ag paste [Fig. 4(a)]. For more rigorous quantitative evaluations, Mg₃Sb₂ thin films were also formed on Si (001) substrates (1–10 Ω cm). The samples (4 × 4 mm² in size) were prepared with vacuum-evaporated Ni/Au electrodes (diameter 0.5 mm) as top contacts. This sample was mounted on a ceramic substrate using Ag paste, and electrical connections to the Ni/Au electrodes were established via gold wire bonding [Fig. 5(a)] to ensure reliable connection and data consistency. For both configurations, an Al electrode was deposited on the backside of the

n-Si substrate. We confirmed that ohmic contacts were formed with In/Mg₃Sb₂ Ni-Au/Mg₃Sb₂, and Al/Si (see the [supplementary material](#)).

For the electrical characterization of the fabricated device, I–V measurements were performed using a source meter (KEITHLEY 2450). The photoresponse was evaluated using 1450 and 1600 nm infrared light-emitting diodes (LEDs) (THORLABS, LED1450L, 3.5 mW and THORLABS, LED1600L, 2 mW). To characterize the fundamental optical properties of the film, transmittance (T) and reflectance (R) spectra were measured using a spectrophotometer (Hitachi High-Tech, UH4150). To verify the photoresponse, a 1550 nm continuous-wave (CW) laser diode (LD) was used as a monochromatic light source in addition to a SWIR-LED in order to confirm the device's response to a specific single wavelength without the broadband influence of the LED. For a more detailed evaluation of the wavelength dependence, a spectral responsivity measurement system was constructed. Infrared light from a SiC light source (Model 80007, ORIEL INSTRUMENTS) was passed through a short-wavelength cut filter (IR-1200, Asahi Spectra) to block light below 1200 nm. The filtered light was modulated by an optical chopper (MC2000B, THORLABS) and introduced into a monochromator (CM110, Spectral Products). The monochromatic light of a specific wavelength was then irradiated onto the device at room temperature. The resulting photocurrent was measured using a lock-in amplifier (Model 7124, SIGNAL RECOVERY) to obtain the spectral responsivity.

Figures 1(a) and 1(b) show the RHEED patterns in the [110]_{Si} direction before and after the growth of the Mg₃Sb₂ thin film at a substrate temperature of 500 °C. The RHEED pattern obtained from the Si(001) substrate before growth exhibits a distinct 2 × 1 surface reconstruction, indicating a clean and well-ordered surface. After the growth process, the RHEED pattern transitioned to streak patterns, suggesting the epitaxial growth of Mg₃Sb₂ on the Si (001) substrate. The red and white arrows correspond to the theoretical diffraction patterns for *c*-plane Mg₃Sb₂ in the direction of $\bar{1}10$ and 110 , respectively. This indicates that a *c*-plane-oriented epitaxial thin film was formed on Si (001) substrates. Figure 1(c) shows the line profile extracted from the regions indicated by the yellow dashed lines in Figs. 1(a) and 1(b). Analysis of the spacing of the diffraction streaks revealed that the Mg₃Sb₂ streaks were slightly narrower than the Si streaks, as indicated by the dotted lines in Fig. 1(c). This observation indicates a larger in-plane lattice parameter for Mg₃Sb₂ than that of the underlying Si substrate, suggesting that the strain in Mg₃Sb₂ was almost relaxed.

Figures 1(d)–1(f) show the RHEED patterns after growth at substrate temperatures of 550 °C, 300 °C, and RT, respectively. The pattern in Fig. 1(d) is similar to that in Fig. 1(a), indicating the absence of Mg₃Sb₂ formation. In fact, the color of the substrate did not change after deposition. This is attributed to the high vapor pressure of Mg, which leads to rapid re-evaporation upon adsorption at high substrate temperatures. Figure 1(e) shows a tendency toward polycrystalline growth at a reduced substrate temperature of 300 °C. However, the analysis of the spot pattern indicated a correspondence with the theoretically calculated diffraction patterns for both Mg₃Sb₂ $\bar{1}10$ and Mg₃Sb₂ 110 , as shown in Fig. 1(b). The blue arrows in Fig. 1(e) indicate diffraction spots of Sb 110 , suggesting the presence of not only Mg₃Sb₂ but also segregated Sb. At RT [Fig. 1(f)], the RHEED pattern exhibits Debye rings, indicating the formation of polycrystalline structures.

Figure 2(a) shows out-of-plane XRD 2θ – ω scans for different substrate temperatures. No Mg₃Sb₂ peaks were observed for the

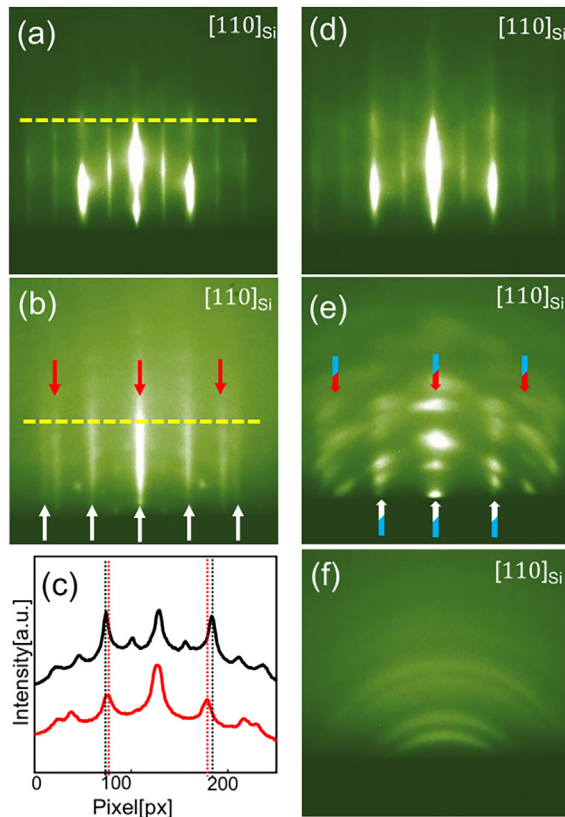


FIG. 1. RHEED patterns in the $[110]_{\text{Si}}$ direction (a) before and (b) after growth at 500°C and (c) line profiles at the yellow dashed lines in the (a) and (b). (d)–(f) after growth at 550°C , 300°C , RT. The red, white, blue, and mixed arrows in (b) and (e) indicate that the red corresponds to the theoretical diffraction pattern of $\text{Mg}_3\text{Sb}_2[110]$, the white to $\text{Mg}_3\text{Sb}_2[110]$, and the blue to Sb $[110]$.

samples grown at RT and 550°C . This absence is attributed to the low crystallinity of the RT sample and the high vapor pressure of Mg at 550°C , as discussed in Fig. 1. In the sample grown at 300°C , three peaks appeared in the range of $2\theta = 23^\circ$ – 26° . The peak at approximately 23° is likely due to the 003 diffraction peak of crystalline Sb, which was also observed in the RHEED patterns. The peak at approximately 24.4° corresponds to the 0002 diffraction peak of Mg_3Sb_2 , whereas the peak at approximately 25.8° corresponds to the 1011 or 0111 diffraction peak of Mg_3Sb_2 . These diffraction peaks were also observed in our previous study on the epitaxial growth of Mg_3Sb_2 on $c\text{-Al}_2\text{O}_3$ substrates, where it was reported that these peaks diminished upon annealing of the substrate.⁴⁶ For the sample grown at 500°C , the intensity of the 1011 or 0111 diffraction peak decreased, and the peak corresponding to crystalline Sb disappeared. Figure 2(b) shows the pole figure of 220_{Si} ($2\theta = 47.37^\circ$). Four-fold symmetric spots were observed near $\chi = 45^\circ$, and since the angle between Si (001) and Si (220) is 45° , these spots were attributed to the Si (001) substrate. Figure 2(c) shows the pole figure of $01\bar{1}2_{\text{Mg}_3\text{Sb}_2}$ ($2\theta = 33.47^\circ$). Twelfold symmetric spots were observed near $\chi = 42^\circ$, and since the angle between $\text{Mg}_3\text{Sb}_2(0001)$ and $\text{Mg}_3\text{Sb}_2(01\bar{1}2)$ is 42.45° , these spots were identified as Mg_3Sb_2 grown with c -plane orientation. This

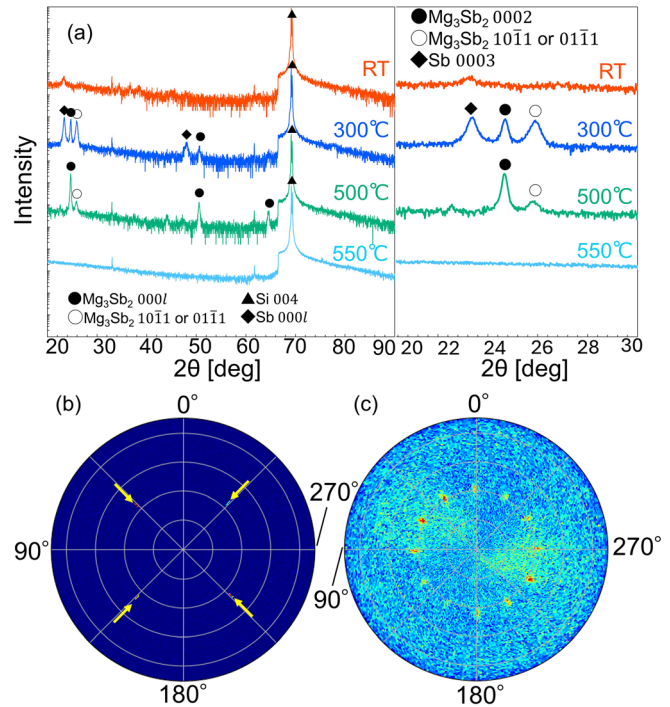


FIG. 2. X-ray diffraction measurements: (a) 2θ - ω scan, (b) 220_{Si} pole figure, and (c) $01\bar{1}2_{\text{Mg}_3\text{Sb}_2}$ pole figure.

observation indicates that Mg_3Sb_2 was epitaxially grown on the Si (001) substrate. The epitaxial relationship between Mg_3Sb_2 and the Si substrate is discussed below.

Figure 3(a) shows a TEM image of Mg_3Sb_2 thin films grown at 500°C . The grains are observed to grow adjacently with diameters of 100–200 nm. Figure 3(b) shows a HAADF-STEM image of the interface regions indicated by a rectangular region in Fig. 3(a). In region A (right side), the atomic arrangement is clearly resolved, whereas it is not discernible in region B (left side), likely because of a slight misalignment of the crystallographic zone axis relative to the electron beam direction. Figures 3(c)–3(f) show selected area electron diffraction (SAED) patterns obtained from the regions indicated in Fig. 3(a). The SAED pattern of the substrate [Fig. 3(c)] is in good agreement with the simulated electron diffraction pattern of Si $[110]$, as shown in Figs. 3(g) and 3(h) (indicated by the black filled circles). Figures 3(d) and 3(e) correspond to the SAED pattern of $\text{Mg}_3\text{Sb}_2[110]$ in Fig. 3(g), while Fig. 3(f) matches that of $\text{Mg}_3\text{Sb}_2[\bar{1}10]$ in Fig. 3(h), which is consistent with the RHEED and pole figure results. Additionally, as shown in Fig. 3(g), the lattice spacing of Si in the $[110]$ direction is in good agreement with that of Mg_3Sb_2 in the $[110]$ direction, which is also confirmed by the SAED pattern at the interface between the substrate and thin film (see the supplementary material). However, the lattice spacing of Si in the $[110]$ direction is not consistent with that of Mg_3Sb_2 in the $[\bar{1}10]$ direction. The lattice constants, calculated from the spacing of the SAED pattern spots in Fig. 3(d), were $a = 0.45$ nm and $c = 0.73$ nm. These values are in close agreement with previously reported data ($a = 0.456$ nm, $c = 0.723$ nm).⁴⁸ Furthermore, the lattice constants for the rotational domain, derived from the pattern in

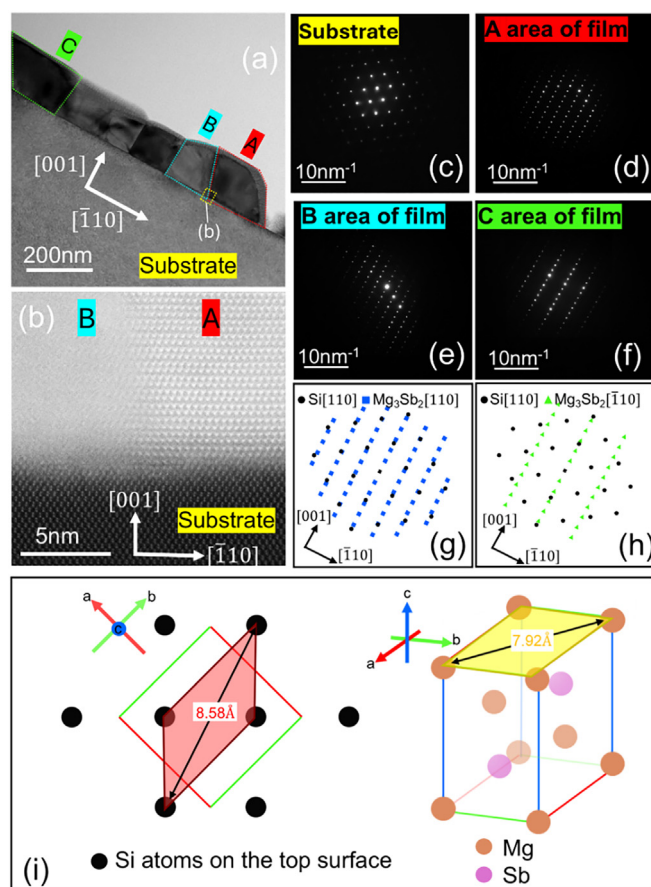


FIG. 3. (a) TEM image and (b) enlarged HAADF-STEM image of the interface between regions A and B. (c)–(f) SAED patterns of the substrate, region A, region B, and region C. (g) Theoretical electron diffraction patterns of Si [110] and Mg_3Sb_2 [110]. (h) Theoretical electron diffraction patterns of Si [110] and Mg_3Sb_2 [110]. (i) Epitaxial relationship between Si (001) and Mg_3Sb_2 (0001).

Fig. 3(f), were $a = 0.45$ nm and $c = 0.74$ nm, indicating no significant difference between the domains.

Based on the above discussion, we propose an epitaxial relationship model, illustrated in Fig. 3(i), which accounts for the observed twelfold symmetry. On the Si(001) surface, the Si atoms in the top atomic layer formed a rhombus shape that aligned with the hexagonal Mg_3Sb_2 unit cell. A comparison of the diagonals of these rhombuses reveals that the spacing is 7.92 Å for Mg_3Sb_2 and 8.58 Å for Si. While a single crystal of Mg_3Sb_2 is expected to show threefold symmetry, the observed twelfold spots in the pole figure can be interpreted as a superposition of diffraction spots from four types of threefold symmetric domains, each rotated by 90° with respect to one another. This domain structure naturally arises from the symmetry mismatch between the threefold symmetric thin film and the fourfold symmetric substrate, providing a physically reasonable explanation for the experimental results. As compared in Table I, Mg_3Sb_2 and Si possess different crystal structures and thermal expansion coefficients. However, the formation of the twelfold symmetric domain structure, arising from the symmetry mismatch between the film and the substrate, effectively

accommodates these differences. This unique domain configuration serves as a relaxation mechanism for the inherent structural and thermal stresses, thereby enabling the heteroepitaxial growth of Mg_3Sb_2 on the Si(001) platform.

Figure 4(a) shows a schematic diagram of the p- Mg_3Sb_2 /n-Si device, and Figure 4(b) shows its current–voltage (I–V) characteristics. The clear rectification behavior indicates the formation of a p–n junction at the heterointerface. The p-type nature of the Mg_3Sb_2 film was independently confirmed. As reported in our previous work,⁴⁶ Hall effect measurements on a similar film grown on an insulating c- Al_2O_3 substrate showed a hole concentration of approximately 10^{17} cm^{-3} . This insulating substrate was chosen to eliminate any potential electrical contributions from the conductive Si substrate during Hall effect measurements.

The infrared photoresponse was then evaluated. Figure 4(c) illustrates the temporal variation of the current under zero bias as the device was periodically illuminated by 1450 and 1600 nm LEDs. The device exhibited distinct photoresponses to both wavelengths at RT without an external voltage. This zero-bias operation is attributed to the photovoltaic effect, which is a desirable characteristic for low-power sensor applications. As seen in the logarithmic plot, the dark current is suppressed to approximately 10^{-11} A, while the photocurrent increases by more than an order of magnitude, resulting in a high ON/OFF ratio. This capability of uncooled, RT operation is a significant practical advantage because it eliminates the need for bulky and costly cooling systems often required for infrared detectors.

To confirm that the photoresponse originated from the Mg_3Sb_2 layer, its optical absorption was directly evaluated, as shown in Fig. 4(d). This spectrum was obtained at RT from an epitaxial Mg_3Sb_2 thin film grown on a transparent c- Al_2O_3 substrate. The thin film was fabricated using a similar method from our recent report, which has been shown to yield high-quality epitaxial films with c-plane orientation on this substrate.⁴⁷ The absorption (A) was simply calculated from the measured transmittance (T) and reflectance (R) using the relation $A = 1 - T - R$ (see the supplementary material). As the raw data exhibit an offset due to scattering from the substrate backside, the spectrum was baseline-corrected by setting the absorption at 0.7 eV to zero, a region where intrinsic absorption is expected to be negligible.

The corrected spectrum in Fig. 4(d) reveals a clear absorption edge beginning around 0.8 eV, confirming that the Mg_3Sb_2 film significantly absorbs light at the operational LED wavelengths of 1450 nm (0.85 eV) and 1600 nm (0.77 eV). A minor absorption feature is also observed around 0.85 eV, corresponding to a known absorption band of water (H_2O) and is attributed to residual water vapor in the instrument's optical path or adsorbed on the sample. In contrast, the Si substrate, with a bandgap of 1.12 eV (detection limit ≈ 1100 nm), is transparent to this long-wavelength light. Therefore, the photoresponse observed in Fig. 4(b) is unequivocally attributed to optical absorption within the Mg_3Sb_2 thin film.

To further investigate the quantitative characteristics of the Mg_3Sb_2 -based photodiode, the spectral response was evaluated at 150 K. As shown in Fig. 5(a), the device exhibits a distinct photosensitivity extending from 1200 nm up to approximately 1600 nm, with a tail reaching 2000 nm. This clearly demonstrates that the device functions as a SWIR detector well beyond the absorption edge of the Si substrate. The quantitative photoresponse was further verified using a 1550 nm laser diode, where a clear linear relationship between the

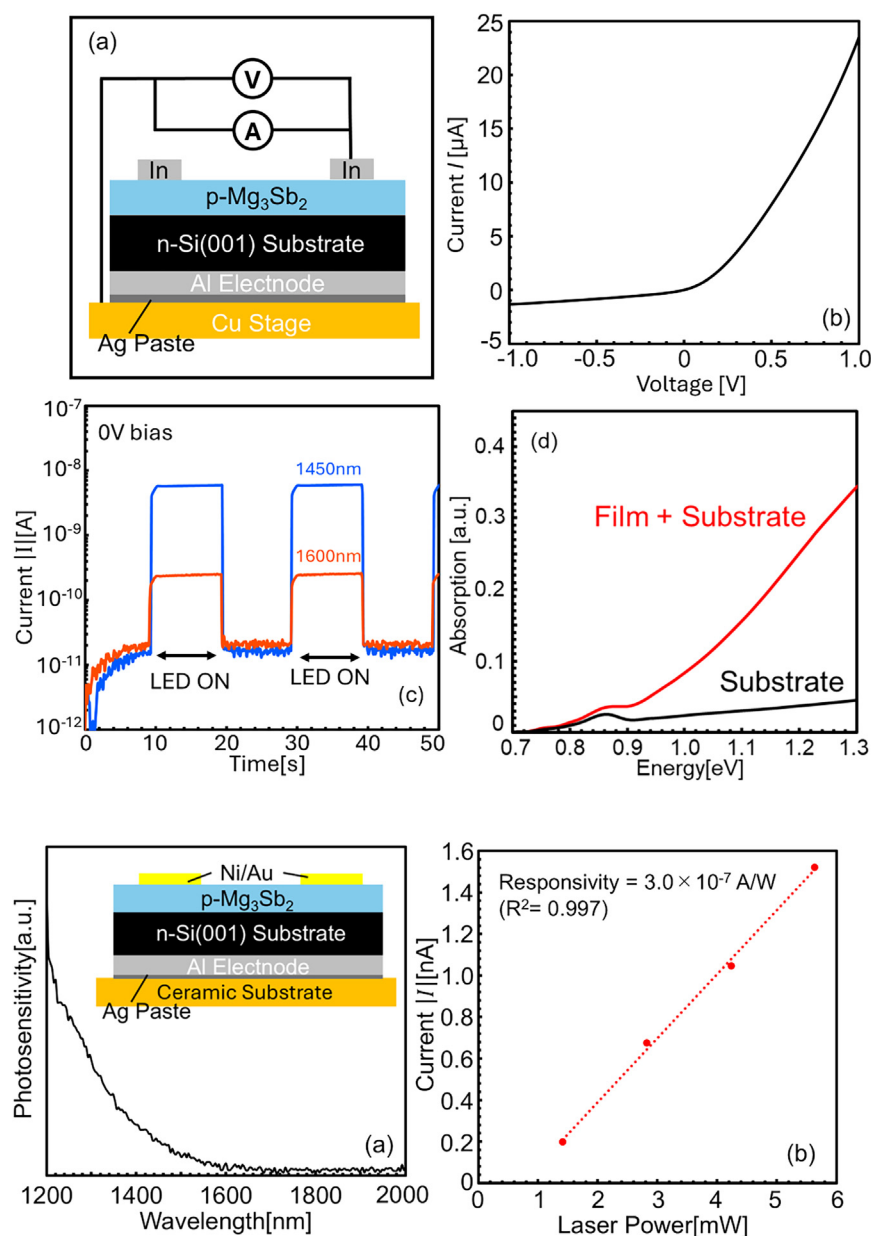


FIG. 4. (a) Schematic diagram of the simplified device structure, (b) I-V characteristics, (c) photoresponse to 1450 and 1600 nm LED modulation at zero bias, and (d) absorption spectra of the Mg₃Sb₂ thin film (red) and the c-Al₂O₃ substrate (black). The spectra have been baseline-corrected by setting the absorbance at 0.7 eV to zero to account for scattering effects.

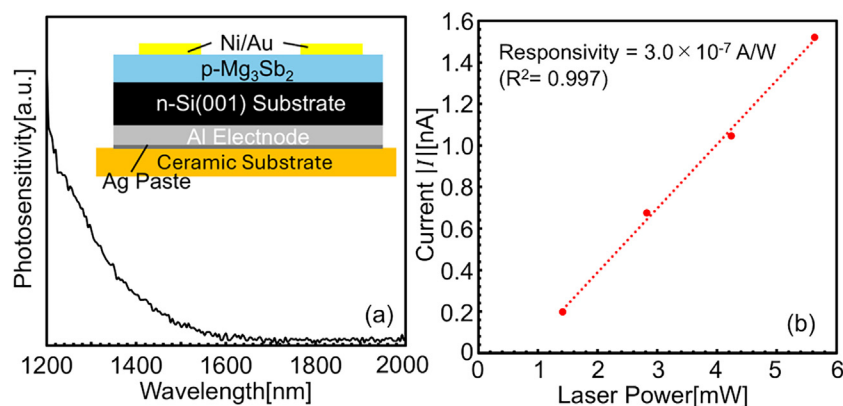


FIG. 5. (a) Spectral photosensitivity measured at 150 K. (b) Photocurrent as a function of incident laser power at 1550 nm.

incident optical power and the generated photocurrent was confirmed, as shown in Fig. 5(b). (The noise spectrum is also shown in the [supplementary material](#).) This linearity indicates that the generated current originates from the photovoltaic effect rather than photothermoelectric effects. These results prove the fundamental potential of epitaxial Mg₃Sb₂ thin films for SWIR detection on Si platforms. However, the current responsivity is relatively low, likely due to the unoptimized crystallinity and device geometry such as film thickness and electrode spacing. In particular, the presence of domain boundaries in the epitaxial Mg₃Sb₂ films on Si(001) may act as recombination centers or carrier traps, which likely accounts for the current requirement of mW-level illumination to generate nA-level photocurrent.

Considering that the temperature-dependent Hall effect measurements of Mg₃Sb₂ thin films on c-Al₂O₃ substrates demonstrated acoustic phonon scattering behavior as a carrier transport mechanism (see the [supplementary material](#)), the crystallinity can be improved through further optimization of the growth method, such as applying an appropriate buffer layer.

In conclusion, we have demonstrated the epitaxial growth of c-axis-oriented Mg₃Sb₂ thin films on Si(001) substrates using the MBE method. The optimal growth temperature was determined to be 500 °C, which yielded high-quality crystalline films. A detailed structural analysis using RHEED, XRD, and TEM revealed a unique epitaxial relationship of Mg₃Sb₂(0001) || Si(001). This is attributed to the

superposition of four types of threefold symmetric domains rotated by 90° with respect to each other. The fabricated p-Mg₃Sb₂/n-Si heterojunction device exhibited clear rectification behavior, confirming the formation of a p–n junction. The device demonstrated a distinct photovoltaic response to short-wavelength infrared light (1450 and 1600 nm) at RT and under zero-bias conditions. Furthermore, evaluation using a 1550 nm laser diode confirmed that the irradiation power dependence of the photocurrent is linear, and spectral sensitivity spectra were obtained at 150 K. These findings establish Mg₃Sb₂ as a promising earth-abundant material for infrared photodetectors that can be integrated with mainstream silicon technology.

See the [supplementary material](#) for the evaluation of electrode contacts, SAED patterns at the interface of Mg₃Sb₂ thin films on Si substrates, optical properties of Mg₃Sb₂ thin films on c-Al₂O₃ substrates, noise characteristics, and carrier transport properties of Mg₃Sb₂ thin films on c-Al₂O₃ substrates.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Nozomu Kiridoshi: Data curation (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Akito Ayukawa:** Data curation (equal); Formal analysis (equal); Writing – review & editing (equal). **Koki Nejo:** Data curation (equal); Investigation (equal). **Takeru Kuriyama:** Data curation (equal); Investigation (equal). **Ryosuke Kobayashi:** Data curation (equal); Investigation (equal). **Wakaba Yamamoto:** Data curation (equal); Formal analysis (equal). **Akira Yasuhara:** Data curation (equal); Formal analysis (equal). **Kohei Sato:** Data curation (equal); Formal analysis (equal). **Haruhiko Udono:** Funding acquisition (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Shunya Sakane:** Conceptualization (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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